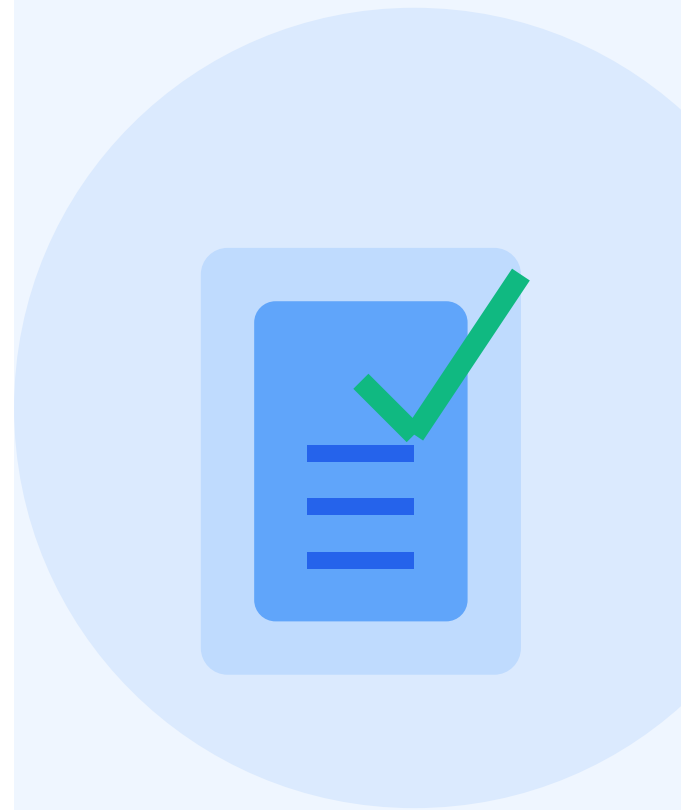


Smart Loan Risk & Approval System

Business Report

End-to-End Machine Learning Application
for Loan Risk Prediction and Approval



Report Summary

Report Type:	Business Report
Generated On:	2026-09-28 21:49
Total Records:	5,000
Total Columns:	21
Target Column:	Loan_Status
Active Model:	Logistic Regression

✓ Data Analysis & Insights

✓ Risk Prediction

✓ Approval Decision

✓ Better Financial Decisions

1. Dataset Overview

Total Records

5,000

Total Columns

21

Numeric Cols

11

Categorical Cols

10

2. Data Quality

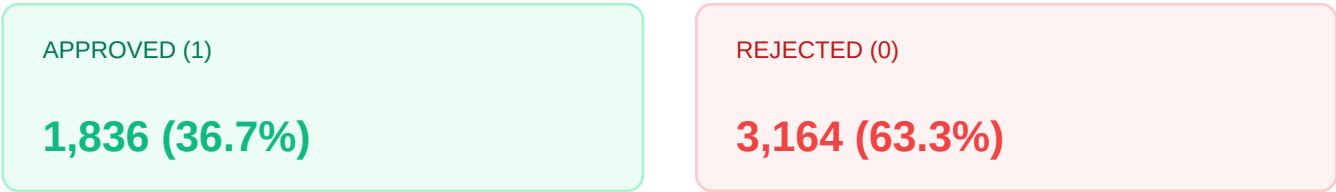
Metric	Count	Percentage
Missing Values	0	0.00%
Duplicate Rows	0	0.00%
Numerical Cols	11	52%
Categorical Cols	10	48%

✓ Dataset is ready for analysis and model training.

3. Missing Values by Column

✓ No missing values detected.
All columns have complete data.

4. Target Analysis (Loan_Status)



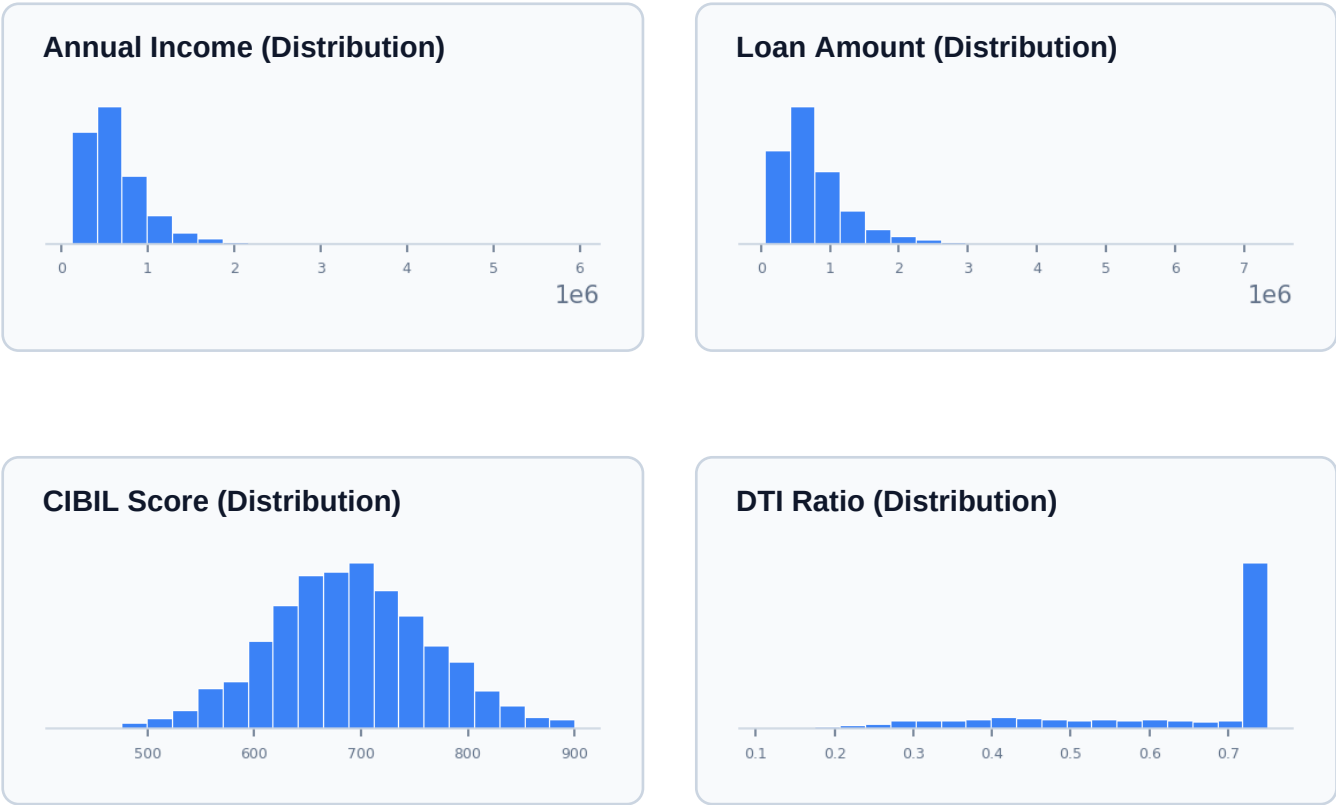
5. Descriptive Statistics

Feature	Mean	Median	Min	Max	Std Dev
Age	42.9	43.0	21.0	65.0	13.0
Employment_Years	15.7	15.1	-0.0	35.0	9.1
Annual_Income	630699.8	540000.0	120000.0	5942000.0	375636.2
Coapplicant_Income	102987.6	65000.0	0.0	1856000.0	140083.2
Loan_Amount	792643.4	638000.0	51000.0	7344000.0	588091.3
Loan_Term_Years	8.8	7.0	2.0	20.0	5.3
Cibil_Score	689.4	689.0	429.0	900.0	75.4
Existing_Loans	1.2	1.0	0.0	6.0	1.1
DTI_Ratio	0.6	0.8	0.1	0.8	0.2
Residential_Assets	2080798.6	1369500.0	66000.0	80846000.0	2521706.4
Bank_Assets	478127.0	343000.0	22000.0	7727000.0	458559.1

6. Outlier Analysis (IQR Method)

Feature	Outlier Count	Outlier %	Severity
Annual_Income	227	4.5%	Medium
Coapplicant_Income	228	4.6%	Medium
Loan_Amount	282	5.6%	Medium
Cibil_Score	43	0.9%	Low
Residential_Assets	372	7.4%	High
Bank_Assets	309	6.2%	Medium

7. Distribution of Key Features



8. Model Training & Comparison

Accuracy

76.7%

Precision

78.6%

Recall

85.4%

F1 Score

81.9%

ROC-AUC

84.1%

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Logistic Regression	0.767	0.786	0.854	0.819	0.841

Only one successfully evaluated model is available.

9. Active Model Details

Model Name

Logistic Regression

Model Version

v20260928214829

Model Type

Loan Risk Classification

Training Status

Ready for Prediction

10. New Applicant Prediction

Applicant Information		Prediction Result	
Age	58.0	MEDIUM RISK	
Employment Years	21.7	Model Confidence: 50.8%	
Annual Income	490000.0	Approval Probability: 50.8%	
Coapplicant Income	126000.0	Risk Probability: 49.2%	
Loan Amount	1174000.0	Decision: REVIEW	
Loan Term Years	10.0		
Cibil Score	689.0		
Existing Loans	1.0		
DTI Ratio	0.75		
Residential Assets	696000.0		
Bank Assets	624000.0		
Gender	Female		
Married	Yes		
Dependents	2		

11. Final Summary

The system analyzed 5,000 historical loan applications using Logistic Regression. The selected model achieved 76.7% accuracy on the evaluation dataset.

Total Records 5,000	Total Columns 21	Selected Model Logistic Regression
Model Accuracy 76.7%	Latest Applicant Result MEDIUM RISK	Approval Probability 50.8%